

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3^r

Mid Term Special Examination, Autumn-2025

Course Code: CSE-2323

Course Title: Digital Logic Design

Time: 1 Hour 30 minutes.

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question No	Question	CLOs	Marks
1.a	Design basic gates (AND, OR, NOT) using only NAND and NOR gates. Show the necessary truth tables, circuit diagrams, and Boolean equations.	CLO1	5
1.b	Design a Full Adder circuit using two Half-Adders and An OR gate and Explain how the Sum and Carry-Out functions of a Full Adder are derived using the XOR operation.	CLO2	5
2.a	Simplify the Boolean function using K-Map: $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ and $d(w, x, y, z) = \sum(0, 2, 5)$	CLO2	4
2.b	Express the Boolean functions: I. $F = A + B'C$ in Sum of Minterm. (SOP) II. $F = xy + x'z$ in a Product of Maxterm.(POS)	CLO1	4
2.c	Using Hamming code (even parity), determine the position of the error in the received 7-bit code word 1011010.	CLO1	2
3.a	Simplify the following Boolean expressions to a minimum number of literals. I. $x'y' + xy + x'y$ II. $ABC + A'B + ABC'$	CLO2	4
OR			
3.a.	Design a 4-bit combinational circuit with inputs A, B, C, D and output F, where $F = 1$ if the number is divisible by 3, otherwise 0. Derive the truth table, simplify the Boolean expression using a K-map, and implement the logic using AND, OR, and NOT gates.	CLO2	4
3.b	For the Boolean functions E and F , as given in the following truth table: I. List the minterms(SoP) and maxterms(PoS) of each function. II. List the minterms(SoP) of $E+F$ and $E \cdot F$. III. Simplify E and F to expressions with a minimum of literals	CLO2	6

X	Y	Z	E	F
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	1	0
1	1	1	0	1

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Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3rd

Mid Term Examination, Spring-2025

Course Code: CSE-2323

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Time: 1 Hour 30 minutes.

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question No	Question	CLOs	Marks
1.a	What do you mean by Parity code? If you receive a hamming code '1101110' with even parity then detect and correct the error.	CLO1	5
1.b	Write down the half adder's truth table and its output Equation. and use AOI gates to implement it. Afterward, change AOI to NAND gates.	CLO2	5
2.a	Simplify the Boolean function $F(A,B,C,D,E) = \sum(0,2,3,4,5,6,7,11,15,16,18,19,23,27,31)$	CLO2	4
2.b	Convert the Boolean expression in canonical form i. $F = AB' + A'BC + A'B'C'D$ (SOP form) ii. $(A' + B).(B' + C + D')(A' + B' + C + D)$ (POS form)	CLO1	4
2.c	Add 8867 with 2832 using BCD addition	CLO1	2
3.a	Simplify and Implement the function F .which has don't care condition d. $F(A,B,C,D) = \sum(0,1,2,3,7,8,10)$ $d(A,B,C,D) = \sum(5,6,11,15)$	CLO2	4
OR			
3.a.	In a room, there are three switches (S1, S2, and S3) controlling a light. The light will turn on (output = 1) under the following conditions: 1. The light turns on if S1 is on (1) and S2 is off (0), regardless of S3. 2. The light turns on if S3 is on (1) and S2 is on (1), regardless of S1. 3. The light turns off (output = 0) in all other cases.	CLO2	4

	Task: Design a combinational logic circuit to control the light, based on the above conditions. Derive the truth table and Simplify it using Karnaugh Map (K-map).		
3.b	An oxygen control system has 4 inputs: • A = Oxygen tank level is low (low=1,otherwise=0) • B = Backup tank available(available=1,not available=0) • C = Patient is critical (critical=1,otherwise=0) • D = Manual override switch (switch on=1,off=0) Oxygen must be supplied (Y = 1) if: • Patient is critical, regardless of anything • OR manual override is on • OR tank is low AND backup is available Tasks i.Derive the truth Table ii.Derive the logical expression of Y iii.Draw the logic circuit	CLO2	6

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE; Semester: 3rd

Mid Term Examination, Spring-2024

Course Code: CSE-2323

Course Title: Digital Logic Design

Time: 1 Hour 30 minutes

Total Marks: 30

Answer the following Three (3) questions. Parts of the same questions must be answered serially. Each question carries 10 marks.

Question : 1	a) Convert Boolean expression in standard form $F=y'+xz'+xyz$. Minimize the following 5 variable SOP function using K map : $F(A,B,C,D,E)=\sum m(0,5,6,8,9,10,11,16,20,24,25,26,27)$. b) Design a circuit that has a 3-bit binary input B_2, B_1, B_0 (where B_2 is MSB and B_0 is LSB) and a single output (Z) specified as follows: $Z = 0$, non prime number $Z = 1$, prime numbers 2, 3, 5, 7	2+5=7 3
Question : 2	a) Simplify the expression using Consensus Theorem $F = (A + B).(A' + C).(B + C)$ b) Design a circuit that the Dual-Function Gate can perform two different logical operations on the data inputs, A and B, depending on the select input X. • $X=0$ NAND • $X=1$ NOR c) Function $F = \sum(1,2,3,4,5,7)$ Minimize and implement with NAND. Is 7421 a self-complementing code or not? Justify. If we have 4 input NAND gate then how many 2 input NAND gates are required to implement it?	2 3 2+1+2=5

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE; Semester: 3rd

Mid Term Examination, Autumn-2023

Course Code: CSE-2323

Time: 1 Hour 30 minutes.

Course Title: Digital Logic Design

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks. Parts of the same questions must be answered serially.

Question : 1	a. Mention the limitations of BCD addition. Define 'Stuck at 0' and 'Stuck at 1'. What are the unique property of Excess-3 code? b. Verify whether 7421 & 3321 are self-complementary code or not. Define the steps associated for converting Binary to Gray code conversion with proper example. Describe Hamming code with proper example. c. If received hamming code is 1110101 with even parity then detect and correct error.	3 1+2+2=5 2
Question : 2	a. Boolean expression to NAND gate implementation: $Y = A' + BC'$ Or a. Minimize the following Boolean functions using K-map with don't care conditions: $F(A, B, C, D, E) = \sum m(1, 2, 5, 9, 13) + \sum d(3, 6, 11, 15)$ b. Write down the steps required for designing a combinational circuit. Design a combinational circuit with four inputs and four outputs. The output generates the 2's complement of the input binary number.	5 5 5
Question : 3	a. Describe 2 bits comparator with proper circuit diagram. Or a. Describe 4 by 2 priority encoder with proper circuit diagram b. What are the advantages and disadvantages of multiplexer? c. Explain decimal to BCD encoder.	5 5 2 3

N.B: The meanings of symbol enclosed in bracket (`) is complement.

****The End****

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Mid Term Examination, Spring-2023

Course Code: CSE-2323

Time: 1 Hour 30 minutes

Course Title: Digital Logic Design

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question : 1	a. Write down the advantages of digital systems over analog systems with proper example. 2 b. Define redundancy theorem with proper example. 2 c. Define Positive and negative logic with truth table. 2 “Excess-3 code is self-complementary code” Is it true or false? Justify your comment. d. Compare between BCD Code and Binary numbers. 1+2=3 Mentioned the rules applied for BCD Addition. e. Just one lines, write down the limitations of BCD Addition. 1	
Question : 2	a. How can we identify whether a code is self-complementary or not? Explain with proper example. 2+1+2=5 Write down major properties of Gray Code. Convert 10110110 Gray Code into Binary Code. Or a. If we have 4 inputs NAND Gate then how many 2 input NAND Gates are required to implement it? 2+3=5 Boolean expressions to NAND Gate implementation: $Y = A\bar{+}BC\bar{}$ ($\bar{}$ means Complement) b. What are the key features of Karnaugh map? Solve the following using Karnaugh Map: 5 $F(A,B,C,D,E) = \sum m(0,1,6,7,8,9,21,22,23,29,31)$	
Question : 3	a. Mention the rules for designing combinational circuit with proper example. 3 4 Or a. What are the necessary conditions for check board configuration? Justify it with Full Adder Circuit. 3 4 b. Construct a Half Adder using NAND gates. 1 3 c. Explain 2 bit comparator with proper circuit diagram. 1 2	

N.B: The meanings of symbol enclosed in bracket ($\bar{}$) is complement.
****The End****

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3rd

Mid Term Examination, Autumn-2022

Course Code: CSE-2523

Time: 1 Hour 30 minutes.

Course Title: Digital Logic Design

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question : 1	a. Design a 4 bit parallel adder using full adders. Define 'Stuck at 0' and 'Stuck at 1'. b. Boolean expression to NOR gate implementation is: $Y = A' + BC'$ Implement the above Boolean expression in terms of AOI. c. How many two input NAND required to implement 4 input AND gate? Minimize the following Boolean functions using K-map: $F(A,B,C,D,E) = \sum m(0,1,6,7,8,9,21,22,23,29,31)$	3+1=4 2 1+3=4
Question : 2	5 bit data 01101 is given. Represent given data in Hamming Code. Or a. If received hamming code is 1110101 with even parity then detect and correct error. b. Design a combinational circuit with three inputs, x, y and z, and the three outputs, A, B, and C, when the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.	5 5 5
Question : 3	a. State Redundancy theorem with proper example. Or a. Define characteristics of positive logic, negative logic and self-dual. b. A logic circuit have three inputs A, B & C. The output F is high when the majority of inputs are logic 1. 1. Minimize the function 2. Implement the circuit c. Using proper algorithm, convert 1011 to Gray code d. State the steps required for designing a combinational circuit with proper example.	3 3 3 3

N.B: The meanings of symbol enclosed in bracket () is complement.

****The End****

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE; Semester: 3rd

Online Mid Term Examination, Spring-2022

Course Code: CSE-2323

Course Title: Digital Logic Design

Time: 03 hour

Total Marks: 21

Answer the following three (3) questions. Each question carries 07 marks. No late submission will be accepted. Students must use gsuite ID for submitting assignment into Google Classroom.

Question : 1	a. Express the Boolean function $F = A + B'C$ as a sum of minterms.	2
	b. A Boolean function F defined on the three input variables x, y and z is 1 if and only if number of 1(one) input is even. Draw the truth table for the above functions and express it in canonical and also minimize the canonical form.	2
	c. We can perform logical operations on strings of bits by considering each pair of corresponding bits separately (called bitwise operation). Given two eight-bit strings $A = 10110001$ and $B = 10101100$, evaluate the eight-bit result after the following logical operations: (a)AND (b) OR (c) XOR (d)NOT A (e) NOT B	1
	d. A Boolean function F defined on the three input variables x, y and z is 1 if and only if number of 1(one) input is even. Draw the truth table for the above functions and express it in canonical and also minimize the canonical form.	2
Question : 2	a. Show that the dual of the exclusive-OR is equal to its complement	1
	b. Write down the necessary conditions for check board configuration	1
	c. Implement the following Boolean function F , together with the don't-care conditions d , using no more than two NOR gates: $F(A, B, C, D) = \sum(2, 4, 10, 12, 14)$ $d(A, B, C, D) = \sum(0, 1, 5, 8).$ Assume that both the normal and complement inputs are available.	2
	d. Derive the circuits for a three-bit parity generator and four-bit parity checker using an odd parity bit.	1
	e. Proof the following using Consensus Theorem: $xy + x'z + yz = xy + x'z$	2

Question : 3	a. Encode the data bits 1101 into the 7 bit even parity Hamming Code. b. Design a combinational circuit with three inputs are A, B, and C, and three outputs are x, y, and z. When the binary input is 0, 1, 2, or 3 the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is two less than the input. c. Implement the Boolean function: $F(A, B, C, D) = C + (A + B')(B' + D)(A' + D)$ with the minimum number of NAND gates. b) NOR gates.	3 2 2
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****The End****